Lab Program-CSA4711-DEEP LEARNING

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1. Image Classification: A deep learning model processes a grayscale image represented as a matrix. Explain how matrix flattening, vector operations, and matrix multiplication can be used to calculate the output of a fully connected neural network layer.

Aim

To write a Python program that demonstrates how a grayscale image represented as a matrix is flattened into a vector and processed using vector operations and matrix multiplication to calculate the output of a fully connected neural network layer.

Procedure

1. Import the NumPy library for matrix and vector operations.
2. Represent a grayscale image as a 2×2 matrix containing pixel values.
3. Flatten the image matrix into a one-dimensional vector.
4. Define the weight matrix and bias vector for the fully connected neural network layer.
5. Perform matrix multiplication between the weight matrix and the flattened image vector.
6. Add the bias vector to the result.
7. Apply the ReLU activation function to obtain the final output.
8. Display the original image, flattened vector, and layer output.

Program

```
from PIL import Image
import numpy as np

# Step 1: Load the picture
img = Image.open(r"C:\PROGRAM\Open CV\zara_003.png").convert("L")

# Step 2: Resize image to 2x2 pixels
img = img.resize((2, 2))

# Step 3: Convert image into a matrix
image_matrix = np.array(img)
```

```
print("Grayscale Image Matrix:")
print(image_matrix)

# Step 4: Flatten matrix into a vector
x = image_matrix.flatten()
print("\nFlattened Image Vector:")
print(x)

# Step 5: Define weights and bias
weights = np.array([
   [0.1, 0.2, 0.3, 0.4],
   [0.4, 0.3, 0.2, 0.1]
])
bias = np.array([1, 2])

# Step 6: Matrix multiplication and bias addition
z = np.dot(weights, x) + bias
print("\nOutput Before Activation:")
print(z)

# Step 7: Apply ReLU activation
output = np.maximum(0, z)
print("\nFinal Layer Output:")
print(output)
```

Intput



Output

```
>>>
================ RESTART: C:/PROGRAM/Deeplearning/lab_dl/dl_ex1.py ==============
Grayscale Image Matrix:
[[ 99 134]
 [ 77 133]]

Flattened Image Vector:
[ 99 134  77 133]

Output Before Activation:
[114.  110.5]

Final Layer Output:
[114.  110.5]
>>>
```